14	**Answer: B**
	θ is the angle between two third order maxima, n = 3 Hence, the angle from the horizontal to the third order maxima is $\frac{\theta}{2}$ $\therefore d\sin\frac{\theta}{2} = 3\lambda \Rightarrow \theta = 2\sin^{-1}\left(\frac{3\lambda}{d}\right)$ <u>Option A:</u> $\theta = 2\sin^{-1}\left(\frac{3\times 450\times 10^{-9}}{\frac{1}{3.2\times 10^{5}}}\right) = 51.189 = 51°\ (2.s.f)$ <u>Option B:</u> $\theta = 2\sin^{-1}\left(\frac{3\times 450\times 10^{-9}}{\frac{1}{1.9\times 10^{5}}}\right) = 29.724 = 30°\ (2.s.f)$ <u>Option C:</u> $\theta = 2\sin^{-1}\left(\frac{3\times 650\times 10^{-9}}{\frac{1}{3.2\times 10^{5}}}\right) = 77.217 = 77°\ (2.s.f)$ <u>Option D:</u> $\theta = 2\sin^{-1}\left(\frac{3\times 650\times 10^{-9}}{\frac{1}{1.9\times 10^{5}}}\right) = 43.492 = 43°\ (2.s.f)$